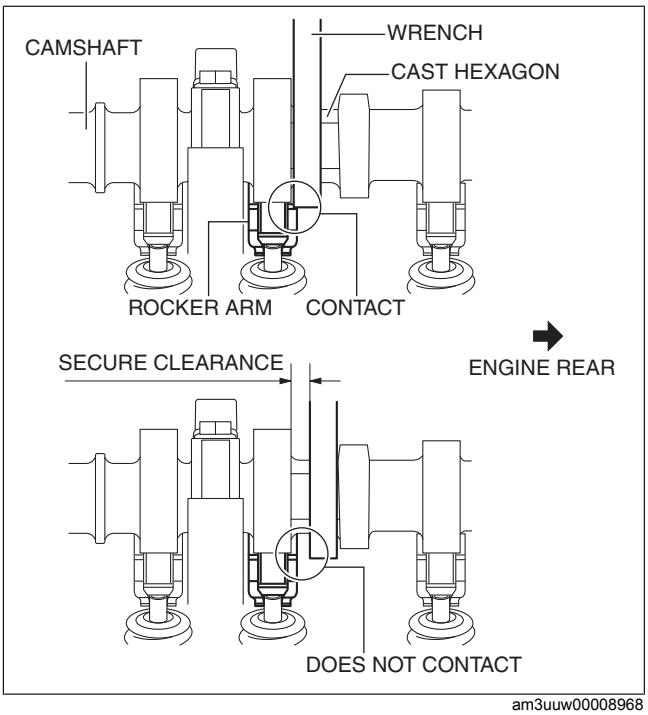
CYLINDER HEAD ASSEMBLY (II)

id011000504400

Caution

- **If the camshaft is rotated with the timing chain removed and the piston at the top dead center position, the valve may contact the piston and the engine could be damaged. When rotating the camshaft with the timing chain removed, rotate it after lowering the piston from the top dead center position.**
- **When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use a wrench on the rear side of the engine as shown in the figure to secure a clearance between the cam.**

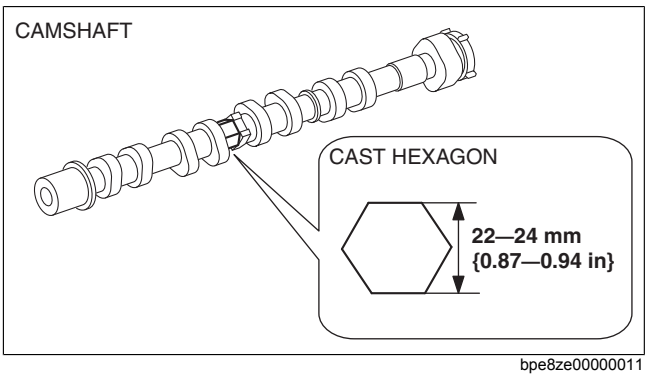


am3uuw00008968

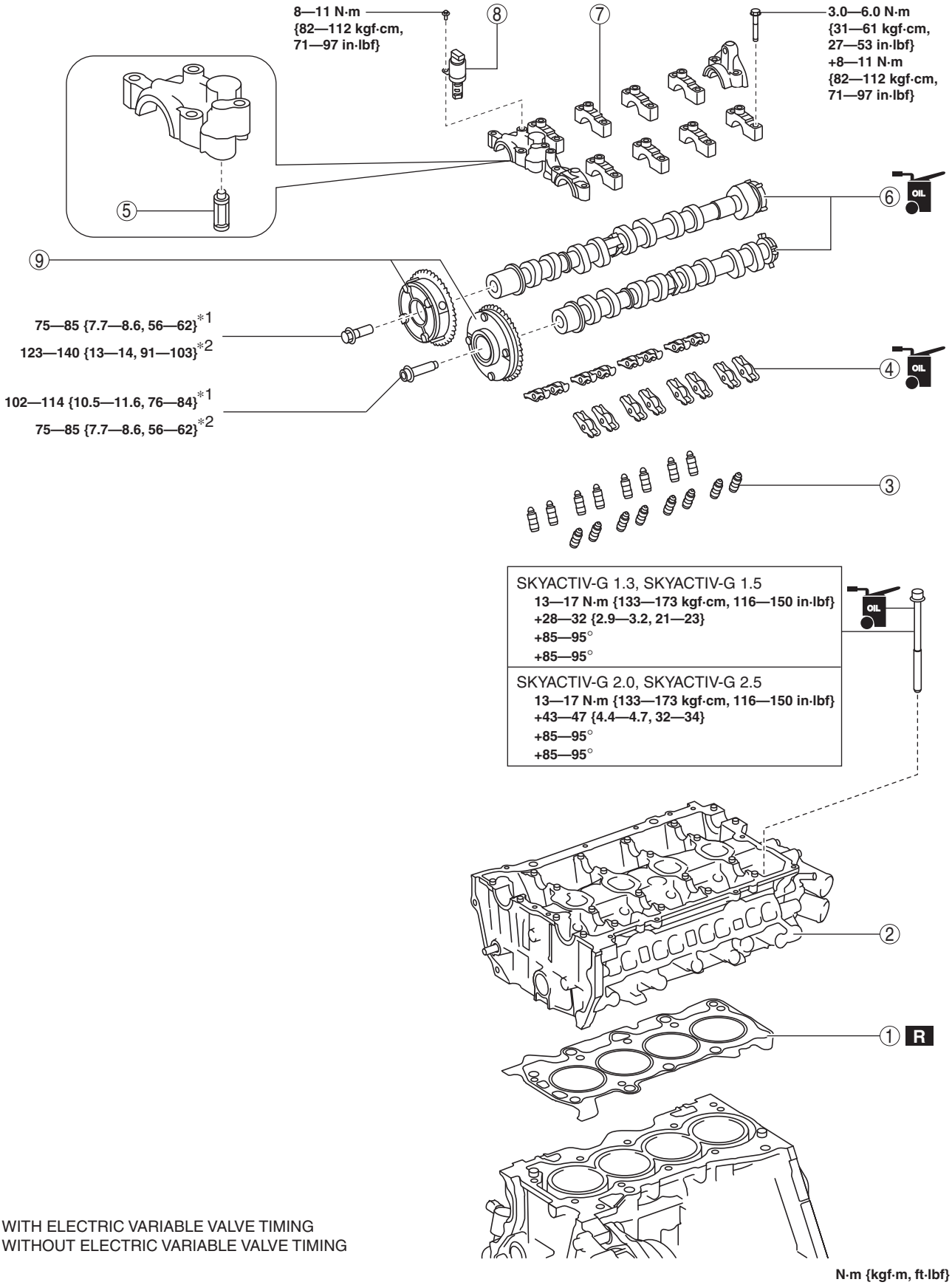
Note

- Width at the cast hexagon of the camshaft is **22—24 mm {0.87—0.94 in}**.

1. Assemble in the order indicated in the table.



bpe8ze00000011



bpe7ze00000015

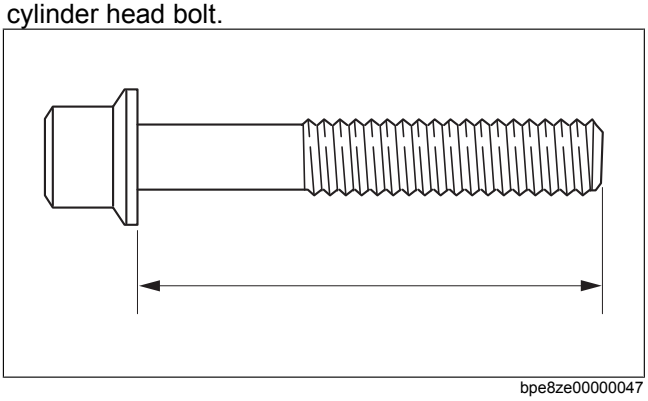
1	Cylinder head gasket
2	Cylinder head (See Cylinder Head Assembly Note.)
3	HLA (See HLA Assembly Note.)
4	Rocker arm (See Rocker Arm Assembly Note.)
5	OCV oil filter
6	Camshaft (See Camshaft Assembly Note.)
7	Camshaft cap (See Camshaft Assembly Note.)
8	OCV
9	**With electric variable valve timing** Electric variable valve timing actuator, hydraulic variable valve timing actuator (See Electric Variable Valve Timing Actuator, Hydraulic Variable Valve Timing Actuator Assembly Note.) **Without electric variable valve timing** Hydraulic variable valve timing actuator, exhaust camshaft sprocket (See Hydraulic Variable Valve Timing Actuator, Exhaust Camshaft Sprocket Assembly Note.)

Cylinder Head Assembly Note

1. Measure the length of the cylinder head bolt
   - If it exceeds the maximum specification, replace the cylinder head bolt.

   Standard cylinder head bolt length
   145.2—145.8 mm {5.717—5.740 in}

   Maximum cylinder head bolt length
   146.5 mm {5.768 in}


bpe8ze00000047

2. When a cylinder head bolt is reused, apply engine oil to any part of the following:
   - Bolt seating surface
   - Cylinder head seating surface

   Caution
   - **Completely remove oil, dirt, and sealant adhering to the cylinder head gasket contacting surfaces. Otherwise, a sealing malfunction may occur.**

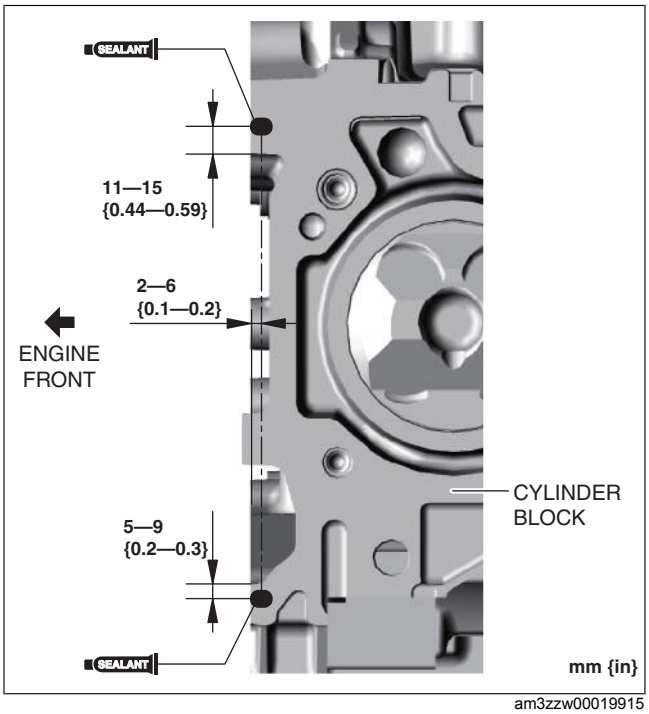
3. Completely remove any oil, dirt, and sealant adhering to the cylinder block.

   Caution
   - **Be aware of the following points, otherwise a sealing malfunction may occur due to hardening of the sealant.**
     - **— Set cylinder head on cylinder block within 10 min after silicone sealant is applied**
     - **— After setting cylinder head, tighten cylinder head bolts immediately**

4. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (Except MX-5 (ND))

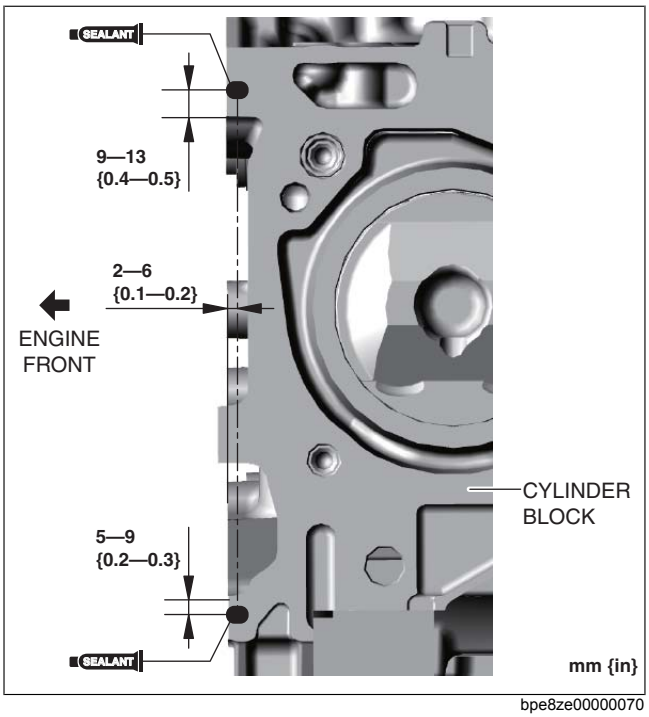
Silicone sealant application diameter
5—10 mm {0.2—0.3 in}



am3zzw00019915

SKYACTIV-G 1.5 (MX-5 (ND))

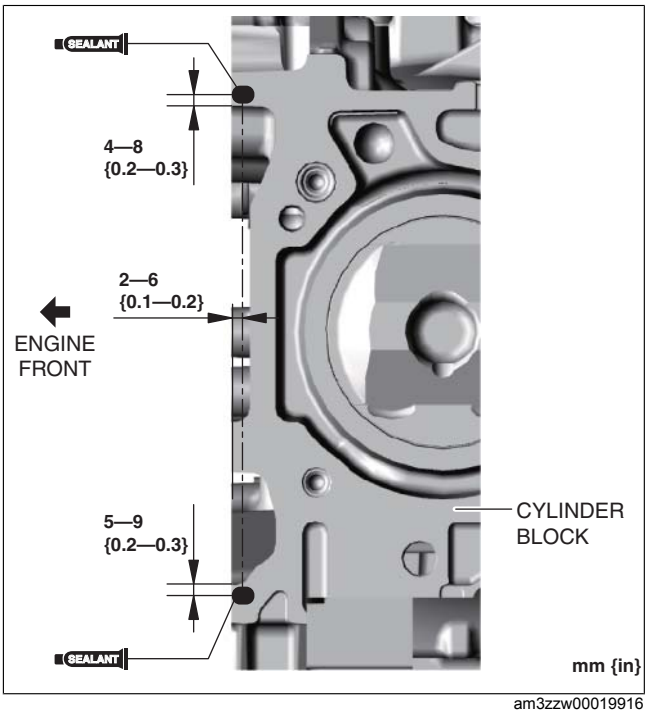
Silicone sealant application diameter
5—10 mm {0.2—0.3 in}



bpe8ze00000070

SKYACTIV-G 2.0

Silicone sealant application diameter
5—10 mm {0.2—0.3 in}

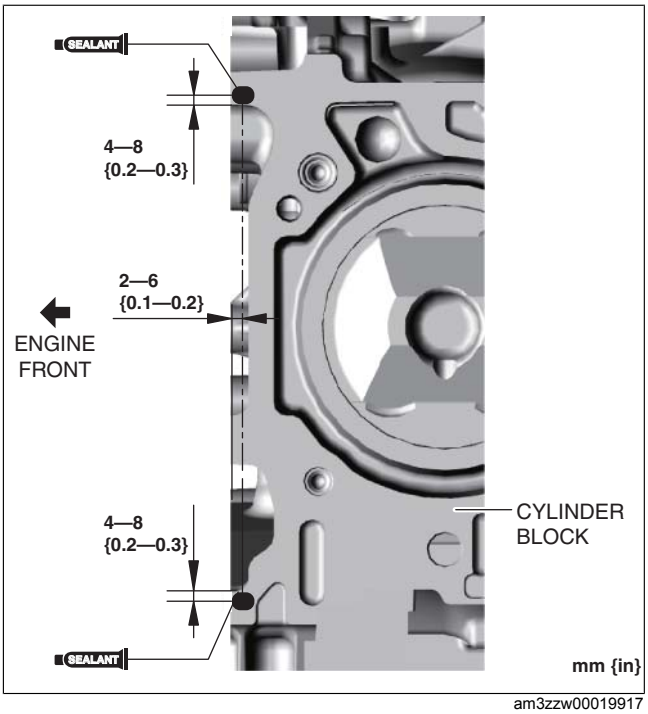


am3zzw00019916

SKYACTIV-G 2.5

Silicone sealant application diameter
5—10 mm {0.2—0.3 in}

5. Install a new cylinder head gasket to the cylinder block.

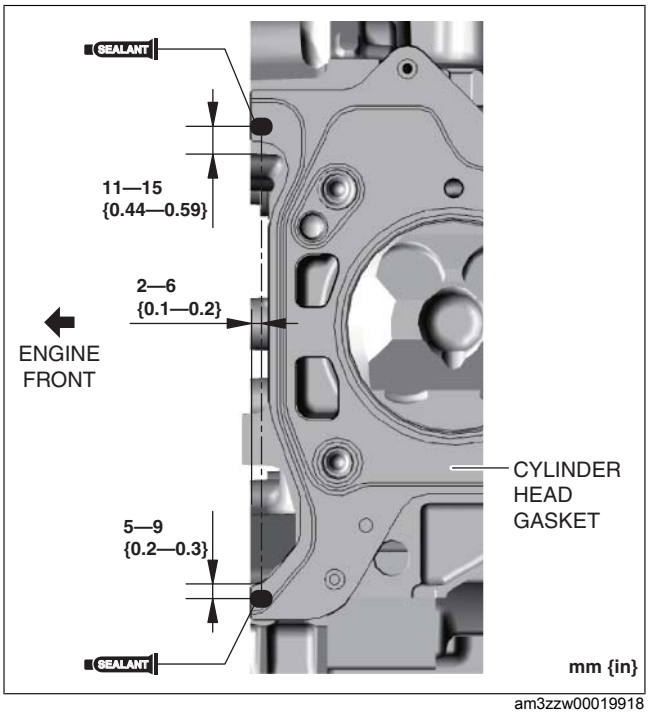


am3zzw00019917

6. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (Except MX-5 (ND))

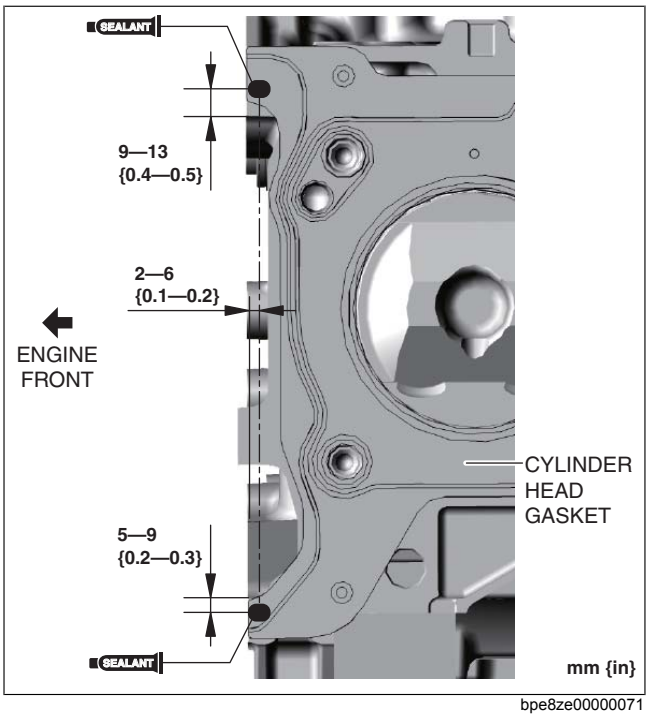
Silicone sealant application diameter
5—10 mm {0.2—0.3 in}



am3zzw00019918

SKYACTIV-G 1.5 (MX-5 (ND))

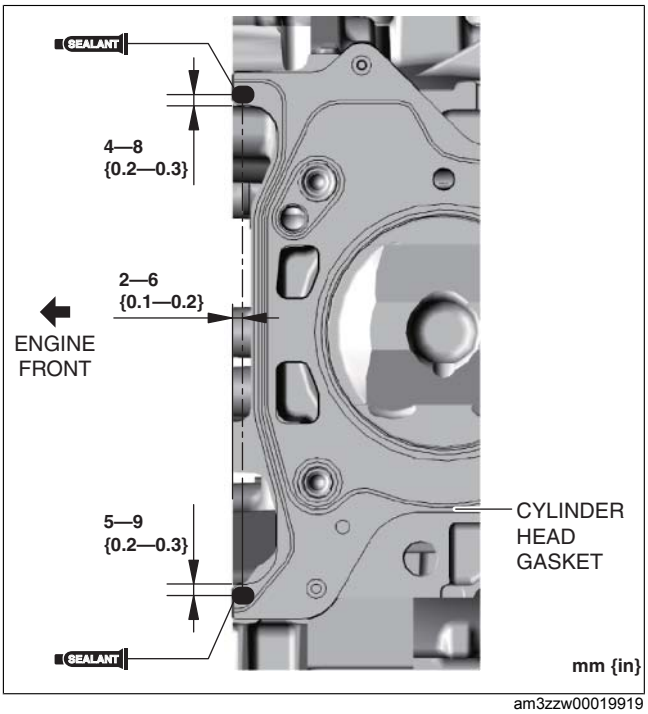
Silicone sealant application diameter
5—10 mm {0.2—0.3 in}



bpe8ze00000071

SKYACTIV-G 2.0

Silicone sealant application diameter 5—10 mm {0.2—0.3 in}

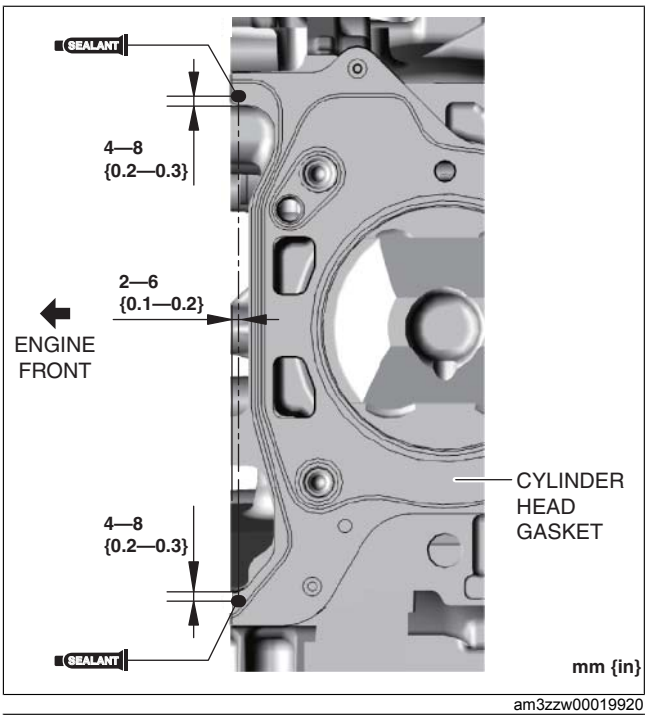


am3zzw00019919

SKYACTIV-G 2.5

Silicone sealant application diameter 5—10 mm {0.2—0.3 in}

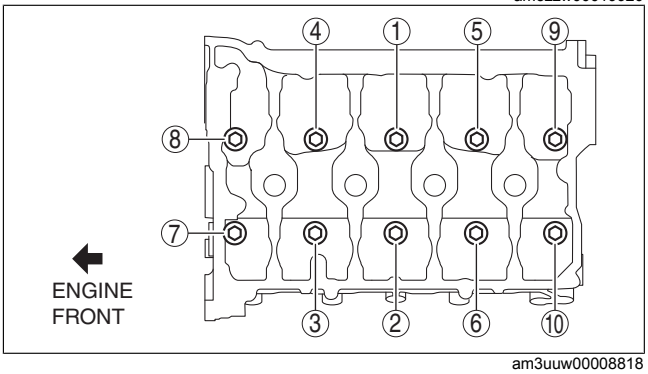
7. Set the cylinder head on the cylinder block.



am3zzw00019920

8. Tighten the cylinder head bolts in the order shown in the following four steps.

Tightening procedure [SKYACTIV-G 1.3, SKYACTIV-G 1.5]
Step 1: 13—17 N·m {133—173 kgf·cm, 116—150 in·lbf}
Step 2: 28—32 N·m {2.9—3.2 kgf·m, 21—23 ft·lbf}
Step 3: 85—95°
Step 4: 85—95°



am3uuw00008818

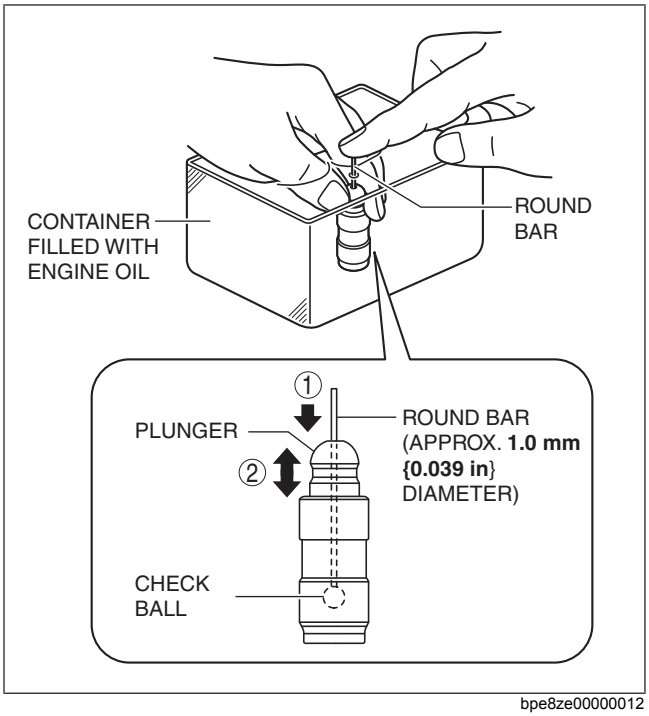
Tightening procedure [SKYACTIV-G 2.0, SKYACTIV-G 2.5]
Step 1: 13—17 N·m {133—173 kgf·cm, 116—150 in·lbf}
Step 2: 43—47 N·m {4.4—4.7 kgf·m, 32—34 ft·lbf}
Step 3: 85—95°
Step 4: 85—95°

HLA Assembly Note

1. Perform HLA air bleeding using the following procedure:
   (1) Put the HLA in a container filled with engine oil.

   Caution
   - **Do not insert the round bar firmly because the check ball spring force is extremely weak.**

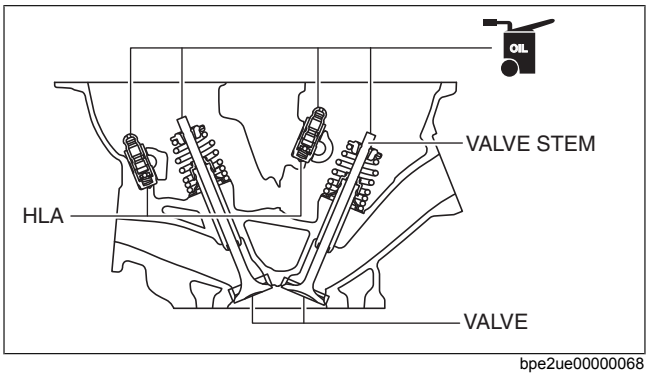
   (2) While lightly pressing the check ball using a round bar (**approx. 1.0 mm {0.039 in}** diameter), bleed air by moving the plunger up and down.
   (3) Press the end of the plunger in the oil and verify that there is no rebounding feel.
      - If rebounding feel cannot be eliminated, replace the HLA.
2. Install the HLAs to their original positions as before removal.



bpe8ze00000012

Rocker Arm Assembly Note

1. Apply engine oil to the HLAs and the end of the valve stems.
2. Install the rocker arms to their original positions as before removal.

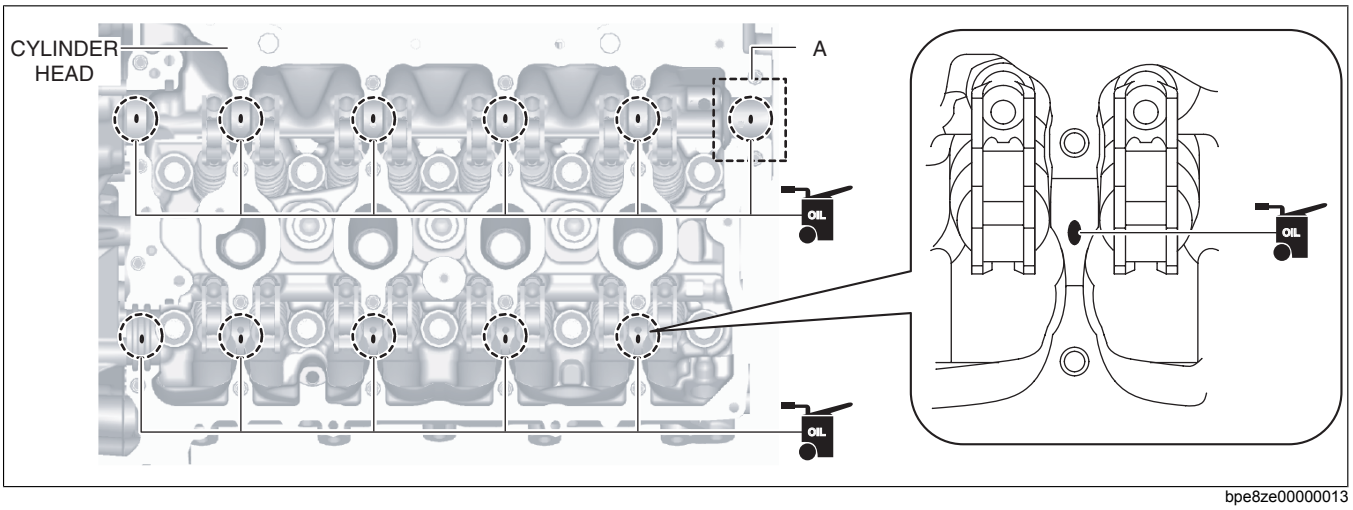


bpe2ue00000068

Camshaft Assembly Note

1. Apply SAE 90 gear oil or equivalent, or engine oil to the positions shown in the figure.

   Caution
   - **Apply 0.05 ml {0.05 cc, 0.003 in^3} or less of oil to area A in the figure.**



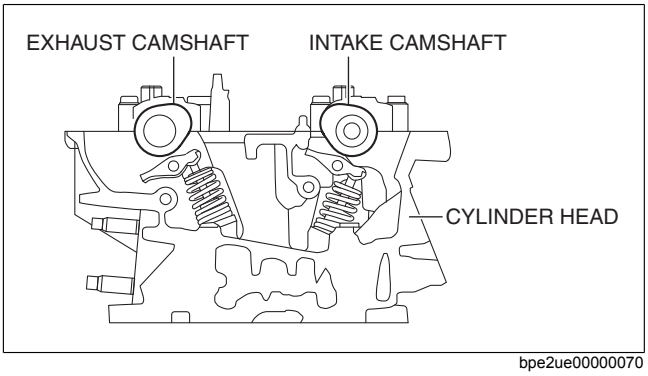
bpe8ze00000013

2. Apply gear oil (SAE 90 or equivalent) or engine oil to the thrust surface (both surfaces front and back) of the front journal on each camshaft.

Caution
- **Be careful not to let oil adheres the engine front cover installation surface. If oil adheres to the engine front cover installation surface, a sealing malfunction may occur when the engine front cover is installed.**
- **If oil adheres to the engine front cover installation surface, remove any oil completely.**

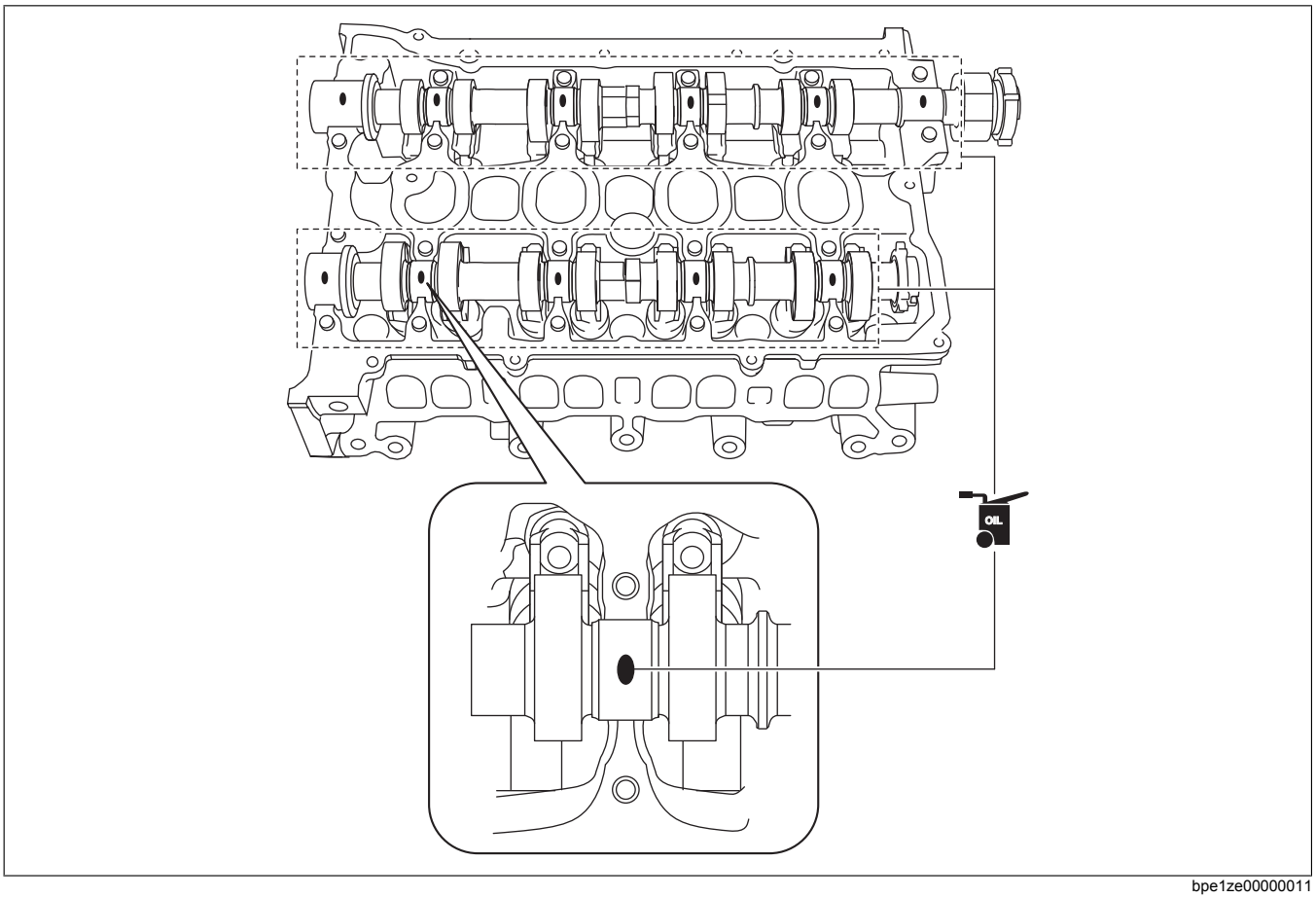
Note
- If oil is applied to the front camshaft cap, oil should not be applied to the thrust surface of the front journal.

3. As shown in the figure, align the cam position of cylinder No.1 around top dead center (TDC) and place the camshafts on the cylinder head.



bpe2ue00000070

4. Apply SAE 90 gear oil or equivalent, or engine oil to the central area of each journal on the camshaft.



bpe1ze00000011

5. Apply SAE 90 gear oil or equivalent, or engine oil to the thrust surface of the front camshaft cap.

Caution
- **Be careful not to let oil adheres the engine front cover installation surface. If oil adheres to the engine front cover installation surface, a sealing malfunction may occur when the engine front cover is installed.**
- **If oil adheres to the engine front cover installation surface, remove any oil completely.**

Note
- If oil is applied to the front journal thrust surface of each camshaft, oil should not be applied to the front camshaft cap.

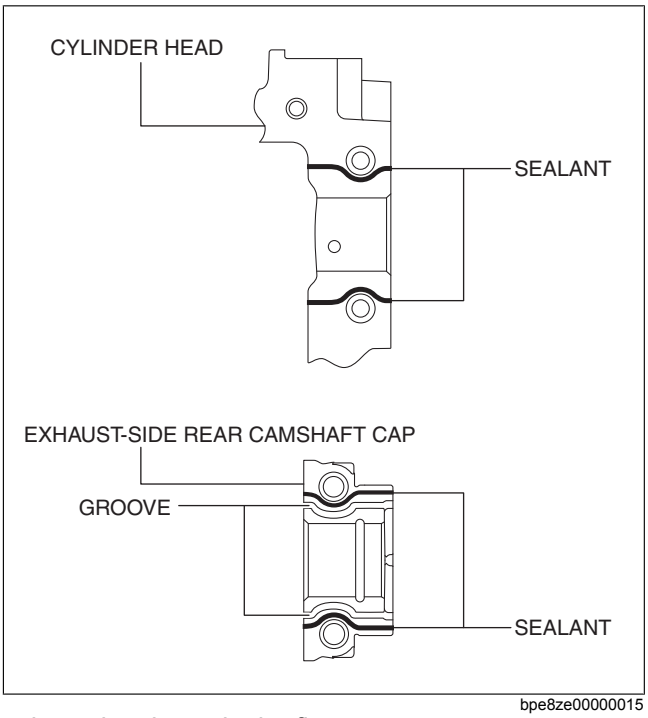
6. Apply sealant (LOCTITE 962T or equivalent) to the rear camshaft cap installation area on the exhaust side of the cylinder head or the rear camshaft cap on the exhaust side.

Note
- To prevent engine oil leakage, apply sealant to the rear camshaft cap installation area on the exhaust side or the rear camshaft caps on the exhaust side of the cylinder head, and seal the journal.

Caution
- **Do not spill sealant on the journal.**

Sealant bead width
1—3 mm {0.04—0.11 in}

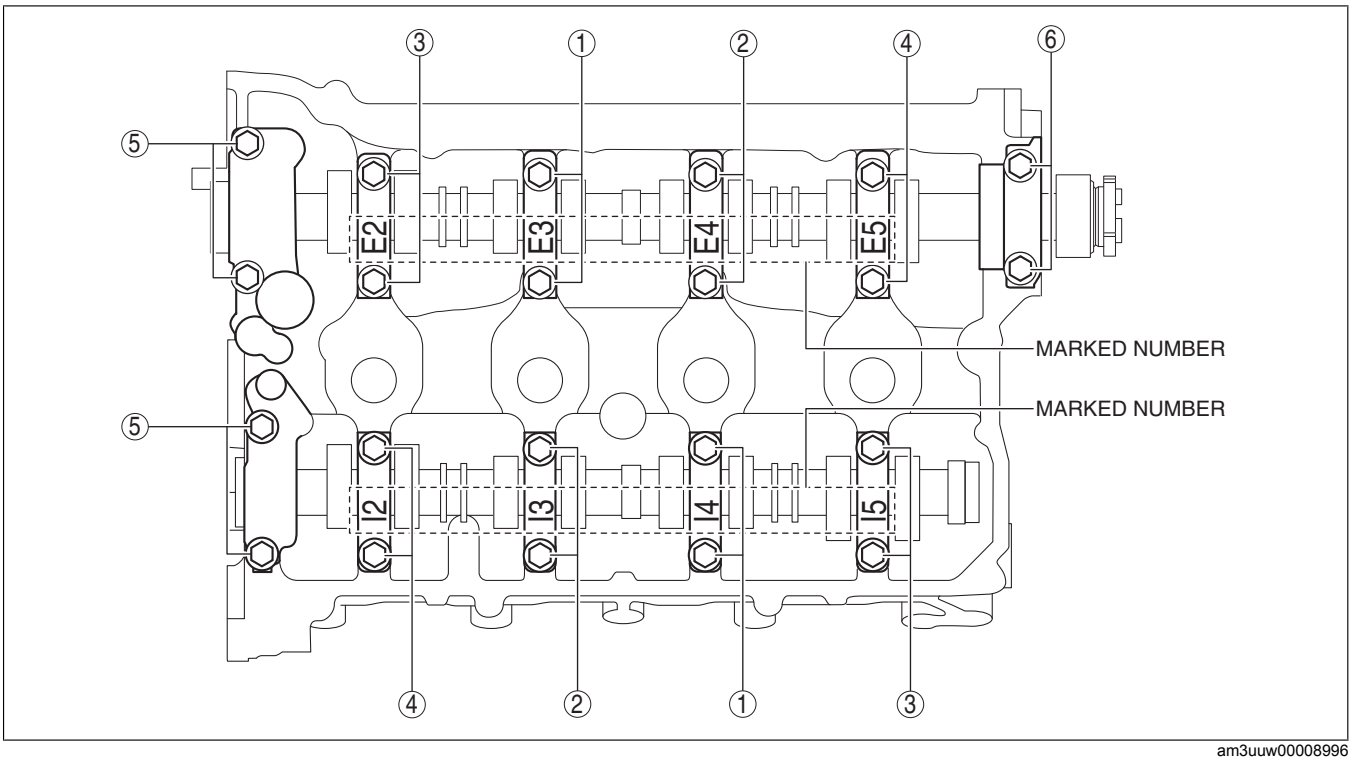
7. Install the camshaft caps in the marked number order, and temporarily tighten the camshaft cap installation bolts in two or three passes evenly.



bpe8ze00000015

8. Tighten the camshaft cap installation bolts in two steps in the order shown in the figure.

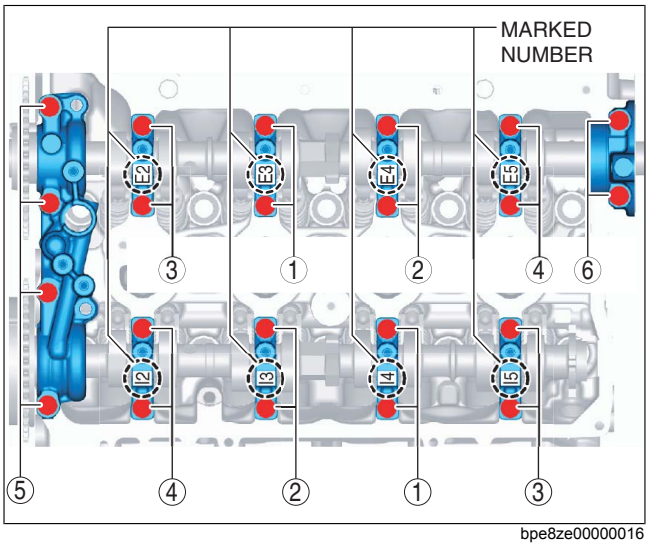
With electric variable valve timing



am3uuw00008996

Without electric variable valve timing

Tightening torque
Step 1: 3.0—6.0 N·m {31—61 kgf·cm, 27—53 in·lbf}
Step 2: 8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

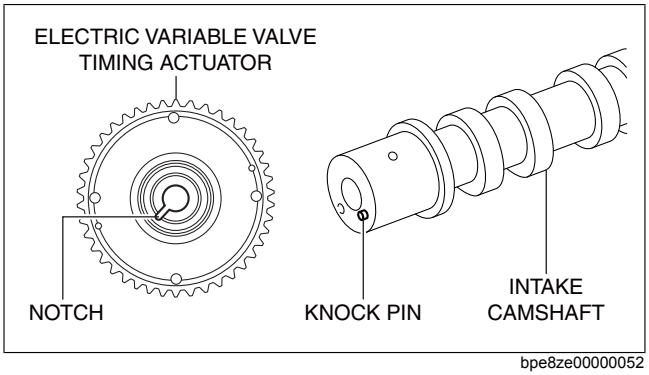


bpe8ze00000016

Electric Variable Valve Timing Actuator, Hydraulic Variable Valve Timing Actuator Assembly Note

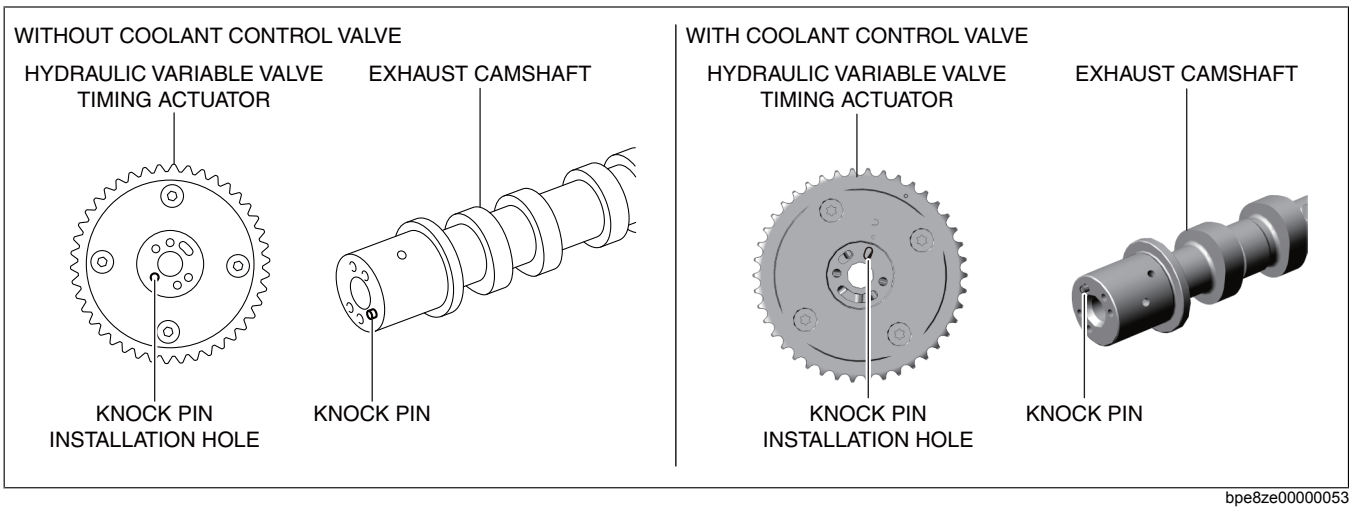
1. Align the knock pin on the end of the camshaft with the notch on the actuator (intake side) or knock pin installation hole (exhaust side), then install the actuator to the camshaft.

Electric variable valve timing actuator



bpe8ze00000052

Hydraulic variable valve timing actuator



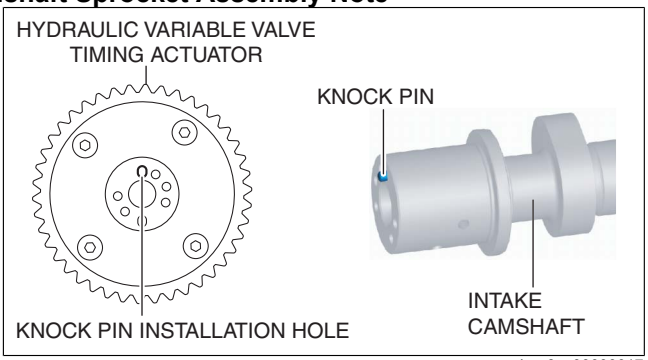
bpe8ze00000053

2. Hold the camshaft using a wrench on the cast hexagon, and tighten the actuator installation bolt.

Tightening torque
Electric variable valve timing actuator (intake side): 102—114 N·m {10.5—11.6 kgf·m, 76—84 ft·lbf}
Hydraulic variable valve timing actuator (exhaust side): 75—85 N·m {7.7—8.6 kgf·m, 56—62 ft·lbf}

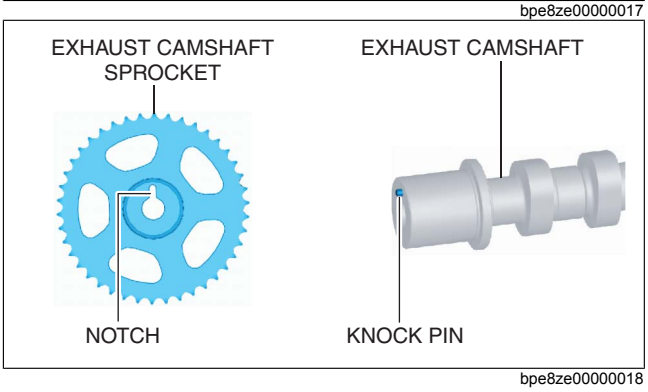
Hydraulic Variable Valve Timing Actuator, Exhaust Camshaft Sprocket Assembly Note

1. Align the knock pin on the end of the intake camshaft with the knock pin installation hole on the actuator side, and temporarily assemble the bolt.



bpe8ze00000017

2. Align the knock pin on the end of the exhaust camshaft with the notch on the exhaust camshaft sprocket side, then assemble the sprocket.
3. Hold the camshaft using a wrench on the cast hexagon, and tighten the bolt.



bpe8ze00000018

Tightening torque

Hydraulic variable valve timing actuator (intake side): 75—85 N·m {7.7—8.6 kgf·m, 56—62 ft·lbf}
Hydraulic variable valve timing actuator (exhaust side): 123—140 N·m {13—14 kgf·m, 91—103 ft·lbf}